

Topic 6: Chemical reactions - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Rates, equilibrium, redox and displacement

Questions in this answer PDF: 18

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 May/June Paper 42 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 16 | Answer source: 0620_s20_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

4(a)

- substance that speeds up a reaction / increases rate (1)
- unchanged (chemically) at the end OR not used up OR lowers activation energy OR provides alternative pathway (1)

4(b)

- rate decreases (1)
- particles further apart / less particles per unit volume (1)
- fewer collisions per unit time / lower collision frequency (1)
- reaction stops because all hydrogen peroxide is used up

4(c)

- steeper gradient (1)
- reaches same volume of oxygen (1)

4(d)

- particles gain kinetic energy / particles move faster (1)
- greater number of collisions with activation energy (or more) / greater number of particles with activation energy (or more) / greater number of particles with energy required for reaction (1)
- more collisions are successful / more collisions are fruitful / more collisions lead to reaction (1)

4(e)

- moles of oxygen
- 48.0
- =
- or moles of oxygen = 0.002 (1)
- moles of hydrogen peroxide = $\times 2$ or moles of hydrogen peroxide = 0.004 (1)
- concentration = $\times 40 = 0.16 \text{ mol / dm}^3$ (1)
- allow ECF

4(f)

- $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
- ALL formulae correct (1)
- balancing (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration $\times$ volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2021 Feb/March Paper 42 Question 3

Main topic: Topic 6: Chemical reactions | Marks: 12 | Answer source: 0620_m21_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

3(a)(i)

- Haber

3(a)(ii)

- reversible / equilibrium

3(a)(iii)

- 450 °C (1)
- 200 atm (1)

3(a)(iv)

- iron / Fe

3(a)(v)

- equilibrium shifts / moves in forwards direction (1)
- fewer molecules / moles (of gas) on RHS (1)

3(a)(vi)

- particles have more energy (1)
- more collisions (between particles) occur per second / per unit time (1)
- a greater percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation
- energy / have energy equal to activation energy (1)

3(b)

- $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$
- $(\text{NH}_4)_2\text{SO}_4$ (1) correct equation (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2021 May/June Paper 41 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 7 | Answer source: 0620_s21_ms_41.pdf

Answer points

4(a)

- the rate of forward reaction equals the rate of the reverse reaction (1)
- concentrations of reactants and products are constant (1)

4(b)(i)

- (increased pressure) nitrogen dioxide particles or molecules (forced) closer together
- OR
- same number of nitrogen dioxide particles or molecules in a smaller volume

4(b)(ii)

- fewer number of gas moles or molecules on left hand side or reactant side (of the equation) opposite reasoning also allowed

4(c)(i)

- shifts to the right

4(c)(ii)

- increase / faster (1)
- increase / faster (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2021 May/June Paper 43 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 7 | Answer source: 0620_s21_ms_43.pdf

Answer points

4(a)

- the rate of forward reaction equals (the rate of the) reverse reaction (1)
- concentrations of reactants and products are constant (1)

4(b)(i)

- same number of gas moles on both sides of the equilibrium / same number of gas molecules on both sides of the equilibrium

4(b)(ii)

- iodine particles or molecules (forced) closer together / same number of iodine particles or molecules in a smaller volume

4(c)(i)

- shifted to the left

4(c)(ii)

- decrease / slower (1)
- decrease / slower (1)
- Question Answer

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2021 Oct/Nov Paper 43 Question 6

Main topic: Topic 6: Chemical reactions | Marks: 10 | Answer source: 0620_w21_ms_43.pdf

Answer points

6(a)

- (a substance that) speeds up a reaction / increases the rate of a reaction (1)
- unchanged chemically at the end (1)

6(b)(i)

- oxygen gas escapes

6(b)(ii)

- rate decreases
- OR
- rate / reaction is fastest at the start (1)
- reaction stops or
- rate = zero (1)

6(c)(i)

- (particles) have more kinetic energy / particles move faster(1)
- more collisions per unit time (1)
- more particles have energy greater than or equal to activation energy
- OR
- more particles have sufficient energy to react
- OR
- greater percentage or greater proportion or greater fraction of collisions have sufficient energy to react
- OR
- greater percentage or greater proportion or greater fraction of collisions have energy greater than or equal to activation energy
- OR
- greater percentage or greater proportion or greater fraction of collisions are successful (1)

6(c)(ii)

- steeper gradient from start (1)
- levels off at the same mass (1)

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Feb/March Paper 42 Question 2

Main topic: Topic 6: Chemical reactions | Marks: 14 | Answer source: 0620_m22_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer points

2(a)

- $\text{MgCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$

2(b)

- eight crosses in second shell of Mg
- 7 dots and 1 cross in third shell of both Cl
- 2+ charge on Mg and - charge on both Cl ions on correct answer line

2(c)

- C atom double bonded to 2 O atoms
- 4 non-bonding e- on each O and no non-bonding e- on C and both octets complete

2(d)(i)

- gradient (of line) decreases

2(d)(ii)

- concentration of particles (of acid) decreases
- lower rate of collisions of particles
- February/March 2022

2(d)(iii)

- a new line steeper than printed line and starts at origin and levels off earlier than printed line
- levels off at the same volume

2(e)

- mol HCl = $0.400 \times 25.0 / 1000 = 0.01(00)$
- mol CO₂ = $/ 2 = 0.0100 / 2 = 0.005(00)$
- volume CO₂ = $\times 24000 = 120 \text{ (cm}^3\text{)}$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2022 Oct/Nov Paper 41 Question 3

Main topic: Topic 6: Chemical reactions | Marks: 14 | Answer source: 0620_w22_ms_41.pdf

Extra topic(s): Topic 5: Chemical energetics, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer points**3(a)**

- Haber (process)

3(b)

- air

3(c)

- reversible

3(d)

- 450 (1)
- 200 (1)

3(e)

- Iron / Fe

3(f)

- reduced temperature:
- (position of) equilibrium moves to right-hand side (1)
- reaction is exothermic (1)
- reduced pressure:
- (position of) equilibrium moves to left-hand side (1)
- more (gaseous) moles on left hand side (1)

3(g)

- rate decreases and particles have less energy (1)
- less collisions (between particles) occur per second / per unit time (1)
- less of the particles/collisions have energy equal to or above the activation energy (1)
- or
- less of the particles / collisions have sufficient energy to react
- or
- a lower percentage / proportion / fraction of collisions (of particles)
- -
- are successful or
- -
- have energy equal to or above activation energy

3(h)

- (NH₄)₂SO₄

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2022 Oct/Nov Paper 43 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 12 | Answer source: 0620_w22_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer points

4(a)(i)

- increases (1)
- decreases (1)
- increases (1)
- increases (1)

4(a)(ii)

- forward reaction is endothermic

4(b)

- 3 dot and cross bonding pairs (1)
- All lone pairs correct.
- All octets should be complete AND dots and crosses should be different on P and all Cl (1)

4(c)

- $\text{P}_4\text{O}_{10} + 6\text{PCl}_5 \rightarrow 10\text{POCl}_3$
- all formula correct and on the correct sides of the arrow (1)
- equation fully correct (1)

4(d)

- H 3.66 / 1
- P 37.80 / 31
- O 58.54 / 16
- OR $\text{H} = 3.66$ $\text{P} = 1.22$ $\text{O} = 3.66$
- OR $\text{H}:\text{P}:\text{O} = 3:1:3$ (1)
- H_3PO_3 (1)

4(e)

- H_3PO_4 (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2023 May/June Paper 41 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 14 | Answer source: 0620_s23_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

4(a)

- increases the rate of reaction /
- speeds up a reaction(1)
- unchanged at the end of the reaction(1)

4(b)(i)

- oxygen escapes from the flask or apparatus

4(b)(ii)

- concentration of hydrogen peroxide is highest at the start / particles of hydrogen peroxide are closest together at the start
- OR
- collision frequency is highest at the start

4(b)(iii)

- the hydrogen peroxide is used up /
- ALL the hydrogen peroxide has reacted or decomposed

4(c)

- kinetic energy of particles increases(1)
- frequency of collisions between particles increases(1)
- more or higher percentage or higher proportion or
- higher fraction of particles have energy greater than / equal to activation energy
- OR
- more of the collisions or higher percentage or higher fraction of collisions have energy greater than or equal to activation energy(1)

4(d)

- $(50.0 \rightarrow 0.200 \div 1000 =) 0.01(1)$
- 0.005(1)
- 0.16(0)(1)

4(e)

- no effect

4(f)

- $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$
- all formulae correct(1)
- equation correct(1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2023 May/June Paper 42 Question 3

Main topic: Topic 6: Chemical reactions | Marks: 17 | Answer source: 0620_s23_ms_42.pdf

Extra topic(s): Topic 10: Chemistry of the environment, Topic 7: Acids, bases and salts

Answer points**3(a)**

- contact (process)

3(b)(i)

- burning sulfur (in air)
- Roasting sulfide ores (in air)

3(b)(ii)

- the rate of forward reaction equals (the rate of the) reverse reaction
- concentrations of reactants and products are constant

3(b)(iii)

- 450 (oC)
- 200 (kPa)
- vanadium(V) oxide

3(b)(iv)

- increases
- increases
- increase
- no change

3(b)(v)

- kinetic energy of particles decreases
- frequency of collisions between particles decreases
- lower percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy
- OR
- fewer of the collisions / particles have energy greater than / equal to activation energy

3(c)

- $4 \rightarrow \blacksquare 2 \text{ or } \blacksquare 8$
- $\text{S} + (4 \rightarrow -2) = -2 \therefore \text{S} = +6$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2023 May/June Paper 43 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 10 | Answer source: 0620_s23_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 7: Acids, bases and salts

Answer points**4(a)(i)**

- concentration (of sulfuric acid particles) decreases
- OR

- frequency of collisions between particles decreases

4(a)(ii)

- all the (sulfuric) acid has reacted

4(b)

- greater surface area (of zinc)
- frequency of collisions between (zinc and acid) particles increases

4(c)

- mol $\text{H}_2\text{SO}_4 = 2.00 \times 25.0/1000 = 0.05(00)$
- mol $\text{H}_2 = 0.05(00)$
- vol $\text{H}_2 = \rightarrow 24 = 1.2(0)$

4(d)(i)

- $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
- ZnCl_2
- equation correct

4(d)(ii)

- lighted splint and (squeaky) pop

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2023 Oct/Nov Paper 41 Question 5

Main topic: Topic 6: Chemical reactions | Marks: 12 | Answer source: 0620_w23_ms_41.pdf

Extra topic(s): Topic 10: Chemistry of the environment

Answer points

5(a)

- the rate of forward reaction equals (the rate of the) reverse reaction (1)
- the concentrations of reactants and products are no longer changing (1)

5(b)(i)

- concentration decreases (1)
- position of equilibrium shifts to left AND fewer moles of gas on left hand side (1)

5(b)(ii)

- concentration increases (1)
- position of equilibrium shifts to the right AND forward reaction is endothermic (1)

5(b)(iii)

- no change (1)
- catalysts do not affect position of equilibrium (1)

5(c)(i)

- carbon dioxide

5(c)(ii)

- greenhouse gases absorb thermal energy (from the Earth) (1)
- and one mark each for any two of:
 - -
 - energy (from the sun) absorbed by Earth's surface
 - -
 - Earth emits or reflects thermal energy
 - -
 - (greenhouse gases) reduces or stops thermal energy loss (into space)
 - -
- increasing in amount of greenhouse gas results in a higher atmospheric temperature

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2024 May/June Paper 41 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 10 | Answer source: 0620_s24_ms_41.pdf

Answer points

4(a)

- The rate of forward reaction equals the rate of the reverse reaction(1)
- concentrations of reactants and products are no longer changing(1)

4(b)(i)

- same number of gas moles on both sides of the equilibrium
- OR
- same number of gas molecules on both sides of the equilibrium

4(b)(ii)

- iodine particles or molecules (forced) closer together
- OR
- same number of iodine particles or molecules in a smaller volume

4(c)

- endothermic

4(d)

- -1 (1)
- zero (1)

4(e)(i)

- copper

4(e)(ii)

- no effect

4(e)(iii)

- (activation energy is) lower

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2024 May/June Paper 43 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 10 | Answer source: 0620_s24_ms_43.pdf

Answer points

4(a)

- (rates are) equal
- (concentrations are) no longer changing

4(b)

- more molecules of nitrogen dioxide per unit volume
- OR
- same number of nitrogen dioxide molecules in a smaller volume

4(c)

- (the forward reaction is) endothermic

4(d)(i)

-
- increases
- no
- change
-
- increases
-
- increases

4(d)(ii)

- (the oxidation number of sulfur in S) = zero
- (the oxidation number of sulfur in SO₃) = +6

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2024 Oct/Nov Paper 42 Question 3

Main topic: Topic 6: Chemical reactions | Marks: 19 | Answer source: 0620_w24_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points

3(a)

- attraction between molecules
- attraction is weak

3(b)

- $2\text{S} + \text{Cl}_2 \rightarrow \text{S}_2\text{Cl}_2$

3(c)

- S with one dot-cross bonding pair with each Cl
- four non-bonding dots for S
- six non-bonding crosses for each Cl

3(d)

- decrease concentration of S_2Cl_2 / P and / or Cl_2
- increase temperature

3(e)(i)

- the minimum energy that colliding particles must have to react

3(e)(ii)

- E_a

3(e)(iii)

- increases no change
- increases increases
- increases

3(f)

- oxidation number of S goes from +1
- oxidation number of S goes to +2
- oxidation number of Cl (goes from -1) to 0 (in Cl_2)
- oxidation numbers of Cl AND S increase

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2024 Oct/Nov Paper 43 Question 5

Main topic: Topic 6: Chemical reactions | Marks: 15 | Answer source: 0620_w24_ms_43.pdf

Answer points

5(a)(i)

- oxygen gas escapes from the flask

5(a)(ii)

- the hydrogen peroxide is used up
- OR
- all the hydrogen peroxide has reacted or decomposed

5(a)(iii)

- time intervals are too large

5(b)(i)

- kinetic energy of particles increases
- frequency of collisions between particles increases
- more / higher percentage OR higher proportion OR higher fraction of particles have energy greater than / equal to
- activation energy
- OR
- more of the collisions / higher percentage OR higher fraction or higher proportion of collisions have energy greater than or
- equal to activation energy

5(b)(ii)

- graph starts at same mass and has steeper gradient
- levels off at the same mass but earlier

5(c)(i)

- oxidation number of manganese
- + 4

5(c)(ii)

- no change

5(d)(i)

- not a closed system

5(d)(ii)

- forward reaction is exothermic

5(d)(iii)

- (to the) right
- there are fewer gas molecules on the right

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2025 Feb/March Paper 42 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 14 | Answer source: 0620_m25_ms_42.pdf

Extra topic(s): Topic 5: Chemical energetics, Topic 2: Atoms, elements and compounds

Answer points

4(a)

- no reactants or products can enter or leave

4(b)

- enthalpy change (of reaction)

4(c)

- (The value of ΔH is) negative

4(d)

- +105 (kJ / mol)

4(e)

- decreases
- increases
- increases
- no change

4(f)

- Bond energy in breaking bonds
- = $1075 + 240 = (+) 1315$
- Bond energy of $\text{COCl}_2 = 1315 - (-105) = 1420$
- $\text{C}=\text{O} = 1420 - 2 \rightarrow \text{C}-\text{Cl}$
- = $1420 - (2 \rightarrow 340) = (+) 740$

4(g)

- both single bonding dot and cross pairs correct
- double $\text{C}=\text{O}$ bond dot and cross pairs are correct
- Complete diagram is correct
- February/March 2025

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2025 Oct/Nov Paper 43 Question 4

Main topic: Topic 6: Chemical reactions | Marks: 17 | Answer source: 0620_w25_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 7: Acids, bases and salts

Answer points

4(a)(i)

- -
- rates are equal

- -

- concentrations are no longer changing

4(a)(ii)

- decreases
- decreases
- no change

4(a)(iii)

- forward reaction is exothermic

4(b)

- USUAL METHOD
- 2490
- 2670
- -180
- ALTERNATIVE METHOD
- 850
- 1030
- -180

4(c)

- 3 dot and cross bonding pairs
- one lone pair on the N atom (to complete the N octet)
- three lone pairs on each Cl atom (to complete each Cl octet)

4(d)(i)

- solid disappears
- effervescence/bubbling/fizzing

4(d)(ii)

- $0.005 \div 5 \rightarrow 10^{-3}$ (mol)
- $0.0025 \div 2.5 \rightarrow 10^{-3}$ (mol)
- 60 (cm³)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*